AI ACCELERATOR BOOTCAMP

Day 3: From Retrieval Pipeline to Working AI Chatbot

DDS HR Enterprise Chatbot
LlamaIndex + Pinecone + OpenAI + Gradio + Hugging Face

Day 1
Clarity, use case, architecture, and why RAG is needed

Day 2
LlamaIndex processing, embeddings, Pinecone, and retrieval

Day 3
Connect the LLM, build UI, deploy, test, and conclude

Where We Are So Far

A clean transition from business clarity to usable AI application

Day 1 — Clarity

- Defined the DDS HR chatbot use case
- Mapped the enterprise AI architecture
- Introduced RAG as the right pattern
- Aligned on user, problem, and value

Day 2 — Retrieval

- Processed documents with LlamaIndex
- Generated embeddings
- Stored vectors in Pinecone
- Tested retrieval with HR questions

Day 3 — Completion

- Connect retrieval to the LLM
- Create the Gradio chatbot UI
- Deploy on Hugging Face
- Test, reflect, and conclude

Bottom line: today we turn a retrieval pipeline into a working chatbot people can actually use.

What We Will Complete Today

Day 3 is about end-to-end usability, not just one more technical component

Today we will:

- Connect the retrieval pipeline to the LLM
- Generate grounded answers using retrieved HR context
- Add system instructions to shape behavior and trust
- Build a simple chatbot interface with Gradio
- Deploy the app to Hugging Face Spaces

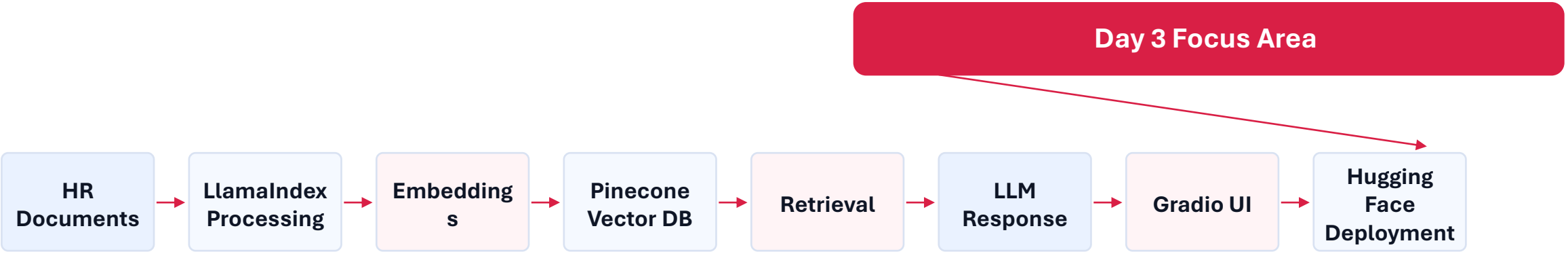
Why this matters:

- This is where the notebook becomes an application
- Users can interact with a real interface
- Deployment makes the work shareable and portfolio-ready
- Testing shows whether the chatbot is useful and reliable

Today is where the project becomes a real AI application.

Complete DDS HR Chatbot Architecture

The full system from enterprise documents to deployed demo



Retrieval + LLM + Interface + Deployment

What Happens When a User Asks a Question

A simple mental model of the end-to-end RAG flow

1. **User asks an HR question**
2. System sends the query through retrieval
3. Pinecone returns relevant HR context
4. Context is passed to the LLM
5. **LLM generates a grounded answer**
6. Answer appears in the Gradio chatbot

Key idea

The LLM does not answer alone. It answers using retrieved enterprise context.

System Instruction and Bot Behavior

Prompting shapes scope, trust, and expected behavior

Bot behavior to enforce

- Answer only HR-related questions
- Use retrieved policy context
- Stay concise, polite, and professional
- Do not invent missing policies
- Say clearly when context is insufficient
- Suggest escalation when needed

Example system instruction

You are a professional HR assistant for DDS.

Answer only using the provided HR and policy context. If the answer is not in the context, say that clearly. Do not invent company rules or policies. Keep answers helpful, concise, and professional. If the query is outside HR scope, ask the user to contact the appropriate team.

Prompting is how we shape behavior, scope, and trust.

Build the Gradio Interface

Make the chatbot usable, interactive, and demo-ready

Why Gradio?

- Fast way to create a working chatbot interface
- Lets users type HR questions and see responses live
- Makes testing easier than running code in a notebook
- Creates a clean demo for peers, managers, or recruiters

DDS HR Enterprise Chatbot

What is the annual leave policy?

A
sk

Employees are entitled to annual leave according to the policy context provided. If you need exact eligibility or timing, please refer to the relevant HR section or contact HR.

Grounded response generated from retrieved HR content

Deploy to Hugging Face Spaces

Move from notebook experiment to a shareable AI demo

Why deploy?

- Share the demo with others quickly
- Showcase your work in a portfolio
- Get feedback from users and peers
- Make the project feel real and complete

Simple deployment flow

- Prepare Gradio app files
- Add requirements and secrets
- Push code to a Space
- Launch and test the public interface

Good practice reminders

- Keep API keys secure
- Verify environment variables
- Test sample HR queries after deployment
- Make sure the UI is clean and usable

Deployment changes the project from a coding exercise into something other people can actually try.

Test the Final Chatbot

A simple testing checklist for the DDS HR assistant

Sample questions to test

- What is the annual leave policy?
- How many sick leave days are allowed?
- Can employees work remotely?
- What is the probation period?
- Who should I contact for HR support?

What to observe

- Is the answer relevant and grounded?
- Does it stay within HR scope?
- Does it avoid guessing when context is missing?
- Is the tone professional and clear?
- Does the chatbot handle edge cases calmly?

Common Day 3 Mistakes

Avoid looking finished without actually being trustworthy

- Connecting the LLM without good retrieved context
- Using weak or vague system instructions
- UI works, but the answer is still not grounded
- Deploying without testing real HR questions
- Focusing on style more than reliability
- Sharing a demo before checking edge cases

A working interface does not automatically mean a trustworthy AI app.

What Participants Should Have by the End

The Day 3 outcome and final build status

Technical output

- A full RAG-based HR chatbot
- Retrieval connected to the LLM
- System instruction in place

Demo output

- A working Gradio interface
- A deployable or deployed Hugging Face Space
- Sample HR tests completed

Learning output

- A better understanding of how enterprise AI is built
- Hands-on experience across the stack
- A project ready to show and improve

Bootcamp Conclusion

From AI Curious to AI Builder in 3 days

AI Curious
Started with interest, questions, and general understanding

→

Use Case Clarity
Defined a concrete DDS HR chatbot problem

→

Retrieval System
Built document search with LlamaIndex and Pinecone

→

Working AI App
Connected the LLM, UI, and deployment path

You did not just learn AI concepts. You built an actual AI application.

Key Takeaways from the Bootcamp

The ideas participants should remember after the final session

What matters most

- Clarity matters before coding
- Enterprise AI needs retrieval, not just prompting
- LlamaIndex helps structure the retrieval workflow
- Pinecone enables semantic search over enterprise data
- System instructions shape reliability and trust

What this means in practice

- A good AI assistant depends on both data and behavior
- A chatbot must be testable, grounded, and usable
- Deployment helps move from prototype to portfolio
- This project is a foundation you can improve over time

If You Had More Time, Improve These Areas

A reflection slide to help participants think like builders

- Better system instructions
- Citations in answers
- More HR documents and scenarios
- Fallback and escalation logic
- Cleaner UI/UX
- Authentication for internal use
- More structured testing
- Analytics and monitoring later

The goal of the bootcamp is a strong prototype, not a perfect enterprise system on day one.

Final Submission Checklist

Keep it simple, practical, and easy to review

What participants should submit

- Bot name
- Which LLM was used
- System instruction used
- Three test questions and outputs
- One key learning from the project
- One improvement if more time were available
- Rough monthly cost estimate
- Hugging Face / Gradio demo link or screenshots

Keep the submission short. Show what you built, tested, and learned.

DECODING
DATA SCIENCE

From AI Curious → AI Builder

The goal was never just to talk about AI.
The goal was to build something real — and now you have.

Keep building. Keep testing. Keep improving.